

CO4-AT-3

MLA0201-FUNDAMENTALS OF MACHINE LEARNING

Annexure A

Comparitive Analysis

Question 1 (20 Marks)

A healthcare organization is developing an intelligent disease diagnosis system using patient symptoms and medical history.

Compare and analyze the suitability of Bayesian Networks and Markov Random Fields (MRFs) for modeling the relationships among diseases and symptoms. Compare both models with respect to graph structure, conditional independence assumptions, inference process, uncertainty handling, computational complexity, and real-world applicability. Justify which model is more appropriate for this healthcare application.

Question 2 (20 Marks)

An autonomous vehicle company is designing a system to recognize driving activities from continuous sensor data collected over time.

Compare and evaluate the performance of Hidden Markov Models (HMMs) and Conditional Random Fields (CRFs) for sequence labeling and activity recognition. Compare the models based on state representation, observation modeling, learning approach, prediction accuracy, handling of contextual features, computational efficiency, and scalability. Recommend the most suitable model for the given application with proper justification.

Question 3 (20 Marks)

A financial analytics company is building an AI-based risk prediction system using customer transaction data and historical financial records.

Compare Directed Graphical Models (Bayesian Networks) and Undirected Graphical Models (Markov Random Fields) for representing probabilistic dependencies in the system. Evaluate both approaches in terms of graph representation, dependency modeling, inference techniques, learning algorithms, generalization capability, computational requirements, and suitability for financial risk prediction. Conclude which graphical model is more effective for this industrial application and justify your recommendation.

1 – BAYESIAN NETWORKS VS MARKOV RANDOM FIELDS

Healthcare Disease Diagnosis System

A healthcare organization can use probabilistic graphical models to represent relationships between diseases, symptoms, and medical history. The two suitable models are Bayesian Networks (BNs) and Markov Random Fields (MRFs).

Comparison

Parameter	Bayesian Network (BN)	Markov Random Field (MRF)
Graph structure	Directed acyclic graph (DAG)	Undirected graph
Dependencies	Represents causal or directional relationships	Represents mutual/symmetric relationships
Conditional independence	Based on d-separation	Based on graph separation
Inference	Variable elimination, belief propagation, sampling	Belief propagation, Gibbs sampling, optimization
Uncertainty handling	Uses conditional probability distributions	Uses potential functions and joint probability
Computational complexity	Can become expensive with complex networks	Inference can also be computationally expensive
Interpretability	Highly interpretable due to directed relationships	Less intuitive for causal interpretation
Healthcare applicability	Very suitable for diagnosis and risk prediction	Useful when relationships are spatial or symmetric

Bayesian Network

A Bayesian Network can represent relationships such as:

Disease → Symptoms → Test Results

For example, diabetes may influence symptoms such as increased thirst and frequent urination. The network can calculate the probability of a disease based on observed symptoms and medical history.

Advantages:

- Represents causal relationships clearly.
- Handles missing and uncertain medical information.
- Supports probabilistic diagnosis.
- Easy to interpret by doctors.

Limitations:

- Requires a directed acyclic structure.
- Designing accurate conditional probability tables can be difficult.

- Inference becomes expensive for very large networks.

Markov Random Field

An MRF represents relationships without specifying a direction. It is useful when symptoms or medical variables have strong mutual dependencies rather than clear causal directions.

Advantages:

- Suitable for complex interdependent variables.
- Does not require directional relationships.
- Can model local relationships effectively.

Limitations:

- Does not naturally represent cause-and-effect relationships.
- Computing the partition function can be expensive.
- Less interpretable for clinical diagnosis.

Critical Evaluation and Insight

For disease diagnosis, the ability to represent cause-and-effect relationships and conditional probabilities is very important. Bayesian Networks provide better interpretability and can answer questions such as *“What is the probability that the patient has disease X given these symptoms?”*

MRFs are more useful when the main objective is to model interacting variables without causal direction, such as spatial medical data or image-based diagnosis.

Conclusion

Bayesian Networks are more appropriate for this healthcare application because they naturally represent disease $\rightarrow$ symptom relationships, handle uncertainty, support probabilistic diagnosis, and provide interpretable results for medical professionals. MRFs can be used as a complementary model when relationships are mainly undirected.

2 – HMM VS CRF

Autonomous Vehicle Activity Recognition

An autonomous vehicle receives continuous sensor information such as speed, acceleration, steering angle, GPS, camera, and LiDAR data. Since driving activities occur over time, sequence models such as Hidden Markov Models (HMMs) and Conditional Random Fields (CRFs) can be used.

Comparison

Parameter	HMM	CRF
State representation	Hidden states represent activities	Labels directly represent activities
Observation modeling	Explicitly models observations	Does not require independent observation model
Learning	Generative learning	Discriminative learning
Prediction accuracy	Good for simple sequential patterns	Usually better with rich features
Contextual features	Limited by independence assumptions	Can use multiple overlapping features
Inference	Forward-Backward, Viterbi	Typically Viterbi/dynamic programming
Computational efficiency	Efficient for simple models	More computationally demanding during training
Scalability	Good for simple state spaces	Better for complex feature-rich problems but training can be expensive

Hidden Markov Model

An HMM assumes that the current activity depends mainly on the previous hidden state, while sensor observations depend on the current state.

Example:

Driving → Braking → Stopping

The HMM can identify the most probable activity sequence using the Viterbi algorithm.

Advantages:

- Simple and computationally efficient.
- Well suited for temporal sequences.
- Easy to implement.
- Works effectively with limited training data.

Limitations:

- Makes strong independence assumptions.
- Observation features are modeled explicitly.
- Difficult to incorporate many correlated sensor features.

Conditional Random Field

A CRF directly models the probability of a sequence of activities given the observed sensor data.

It can use features such as:

- Vehicle speed
- Acceleration
- Steering angle
- Previous activity
- Road conditions
- Time-dependent sensor patterns

Advantages:

- Uses rich and overlapping contextual features.
- Does not require strong independence assumptions about observations.
- Often provides better sequence-labeling accuracy.
- Suitable for multi-sensor activity recognition.

Limitations:

- Training can be computationally expensive.
- Feature engineering may be required.
- Large CRFs can require significant computational resources.

Critical Evaluation and Insight

HMMs are attractive when the vehicle system requires fast and simple real-time processing. However, autonomous driving generates many correlated sensor features, and the independence assumptions of HMMs may reduce accuracy.

CRFs can combine multiple sensor features and contextual information, making them more suitable for recognizing activities such as turning, braking, accelerating, lane changing, and stopping.

Conclusion

CRF is more suitable for the given autonomous vehicle application because it can incorporate rich contextual sensor features and model dependencies between activity labels without strong observation-independence assumptions. Although HMMs are computationally simpler, CRFs generally provide better performance for complex, multi-sensor sequence labeling.

3 – BAYESIAN NETWORKS VS MARKOV RANDOM FIELDS

Financial Risk Prediction System – 20 Marks

A financial analytics company can use graphical models to represent dependencies between transactions, customer behavior, income, credit history, loans, and financial risk.

Comparison

Parameter	Bayesian Network	Markov Random Field
Graph representation	Directed acyclic graph	Undirected graph
Dependency modeling	Directional dependencies	Symmetric dependencies
Causal interpretation	Strong	Limited
Inference	Variable elimination, belief propagation, sampling	Belief propagation, MCMC, optimization
Learning	Structure and parameter learning	Parameter/potential learning
Generalization	Good when causal structure is known	Good for highly interconnected variables
Computational requirements	Can increase with network complexity	Partition-function computation can be expensive
Financial suitability	Strong for risk prediction and decision support	Useful for correlated/interdependent financial variables

Bayesian Network

A Bayesian Network can represent relationships such as:

Income → Loan Eligibility → Loan Repayment → Default Risk

It can calculate the probability of financial risk based on historical records and current customer information.

Advantages:

- Clearly represents dependency direction.
- Supports causal reasoning.
- Handles missing and uncertain data.
- Provides interpretable risk probabilities.
- Useful for decision-making and credit-risk analysis.

Limitations:

- Requires a suitable network structure.

- Conditional probability estimation can be difficult.
- Complex networks may require expensive inference.

Markov Random Field

An MRF models variables as an interconnected network without assigning directions. For example, transaction behavior, spending patterns, account activity, and customer relationships can be modeled as mutually dependent variables.

Advantages:

- Handles strong interactions between variables.
- Suitable when dependencies are not naturally directional.
- Useful for detecting patterns in interconnected financial data.

Limitations:

- Does not naturally represent causality.
- Inference may become expensive for large networks.
- Partition-function calculation can be computationally difficult.

Critical Evaluation

Financial risk prediction often requires understanding why a customer is considered risky, not just identifying correlations. Bayesian Networks provide a clearer representation of relationships such as income affecting repayment ability and repayment behavior affecting default risk.

MRFs can capture complex correlations effectively, but their undirected structure makes them less suitable for explaining causal financial processes.

Synthesis and Insight

The choice depends on the objective:

- Causal risk analysis and explainable decisions → Bayesian Network
- Complex interconnected relationships → MRF
- Probabilistic risk estimation → Bayesian Network
- Correlation-based pattern modeling → MRF

Since financial institutions require interpretable and explainable risk predictions, causal and directional relationships are particularly valuable.

Conclusion

Bayesian Networks are more effective for the financial risk prediction application. They provide interpretable dependency structures, handle uncertainty, support probabilistic prediction, and allow causal reasoning. MRFs are useful for highly interconnected transaction patterns, but Bayesian Networks provide a better balance of interpretability, prediction, and decision support for industrial financial risk analysis.
